

Machine Learning Exercise 1B

Problem 1 *Here you will implement linear regression with multiple variables to predict the prices of houses. The file `ex1data2.txt` contains a training set of housing prices in Portland, Oregon. The first column is the size of the house (in square feet), the second column is the number of bedrooms, and the third column is the price of the house.*

By looking at the values, note that house sizes are about 1000 times the number of bedrooms. You will perform feature normalization to make make gradient descent converge much more quickly.

- i) Load and visualize the data*
- ii) Perform feature normalization*
- iii) Run gradient descent.*
- iv) Check the progress of the gradient descent by checking that $J(\theta)$ is decreasing after each step.*
- v) Use the final values of θ to predict the price of a house with 1650 square feet and 3 bedrooms.*
- vi) Check if you get the same values of θ using some Python library such as for instance `scikit-learn`.*